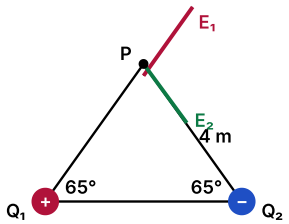
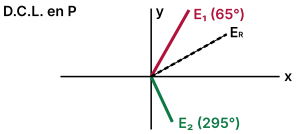
2do parcial · I · Campo en triángulo

E_R y θ en P

$+Q_1 = 7\times10^{-9}$ C (abajo izq.), $-Q_2 = 4\times10^{-9}$ C (abajo der.), lados = 400 cm ÷ 100 = **4 m**, ángulos de base 65°.
E_1 **sale** de $+Q_1$ → 65° · E_2 **entra** a $-Q_2$ → 360 − 65 = 295°



✍ COMO EN LA PIZARRA	📱 CALCULADORA
d = 400 cm = 4 m	400 ÷ 100 = →4
$E_1 = 9\times10^9 N\cdot m^2/C^2\ (7\times10^{-9}C) / (4m)^2 = 3.94$ N/C	9 × 7 →63 · 4 x^2 →16 63 ÷ 16 = →3.94
$E_2 = 9\times10^9 N\cdot m^2/C^2\ (4\times10^{-9}C) / (4m)^2 = 2.25$ N/C	9 × 4 →36 · 36 ÷ 16 = →2.25
$E_1x = E_1\cos65° = 3.94\cos65° = 1.67$ N/C $E_2x = E_2\cos295° = 2.25\cos295° = 0.95$ N/C	65 cos × 3.94 = →1.67 295 cos × 2.25 = →0.95
$E_1y = E_1\operatorname{sen}65° = 3.94\operatorname{sen}65° = 3.57$ N/C $E_2y = E_2\operatorname{sen}295° = 2.25\operatorname{sen}295° = -2.04$ N/C	65 sin × 3.94 = →3.57 295 sin × 2.25 = →−2.04
ΣEx = 1.67 + 0.95 = 2.62 N/C **ΣEy = 3.57 − 2.04 = 1.53 N/C**	1.67 + 0.95 = 3.57 − 2.04 =
$E_R = \sqrt{(2.62)^2 + (1.53)^2}$	2.62 x^2 + 1.53 x^2 = $\sqrt{x}$ →3.03
a) E_R = 3.03 N/C	
$tg\beta = 1.53 / 2.62 = 0.58$ $\beta = tg^{-1}(1.53 / 2.62) = 30.28°$ θ = β (cuadrante I)	1.53 ÷ 2.62 = →0.58 2nd $\tan^{-1}$ →30.28
b) θ = 30.28°	



⚠ Nunca uses 400 directo: la fórmula pide **metros** (k está en $N\cdot m^2/C^2$). Con 400 saldría 0.000394, que está mal.
⚠ sen295° es **negativo** (E_2 apunta hacia abajo). Calculadora en grados: 60 cos = 0.5

2do parcial · II · Distancia entre masas

gravitación

$m_1 = 18$ kg, $m_2 = 24$ kg, $F = 5\times10^{-12}$ N, $G = 6.67\times10^{-11}$ $N\cdot m^2/kg^2$. Hallar d.

✍ COMO EN LA PIZARRA	📱 CALCULADORA
Despeje: $F = G\cdot m_1\cdot m_2/d^2$ $d^2\cdot F = G\cdot m_1\cdot m_2$ $d^2 = G\cdot m_1\cdot m_2 / F$ $d = \sqrt{(G\cdot m_1\cdot m_2 / F)}$	
$d = \sqrt{(6.67\times10^{-11} N\cdot m^2/kg^2)(18kg)(24kg) / 5\times10^{-12} N}$	
$d = \sqrt{(6.67\times10^{-11} N\cdot m^2/kg^2)(432kg^2) / 5\times10^{-12} N}$	18 × 24 = →432
$d = \sqrt{(2881.44\times10^{-11} / 5\times10^{-12})}\ m^2$ $10^{-11+12} = 10^1$	6.67 × 432 = →2881.44 ÷ 5 = →576.288
$d = \sqrt{(576.288\times10)} = \sqrt{5762.88}\ m^2$	× 10 = →5762.88 · $\sqrt{x}$ →75.91
d = 75.91 m	